

Data Mining - Clustering

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Minkowski Distance

$X = (x_1, x_2, \dots, x_n)$ and $Y = (y_1, y_2, \dots, y_n) \in \mathbb{R}^n$

$$D(X, Y) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{\frac{1}{p}}$$

p=1 : Manhattan Distance

p=2 : Euclidean Distance



Minkowski Distance

Example

$A(0,0)$, $B(3,0)$, $C(3,4)$:

Manhattan Distance:

$d(A,B)=3$, $d(B,C)=4$, $d(A,C)=7$

Euclidean Distance:

$d(A,B)=3$, $d(B,C)=4$, $d(A,C)=5$



Jaccard Similarity Coefficient

Jaccard similarity coefficient (Jaccard index)

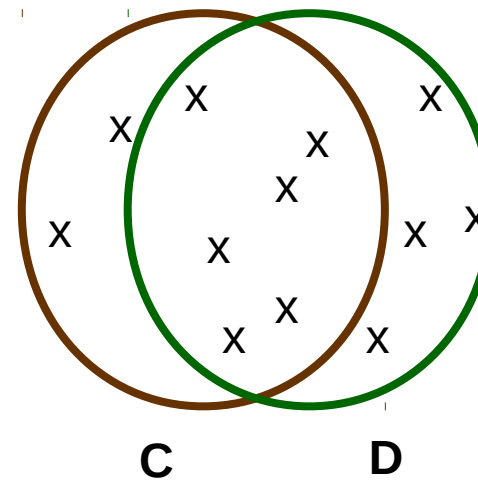
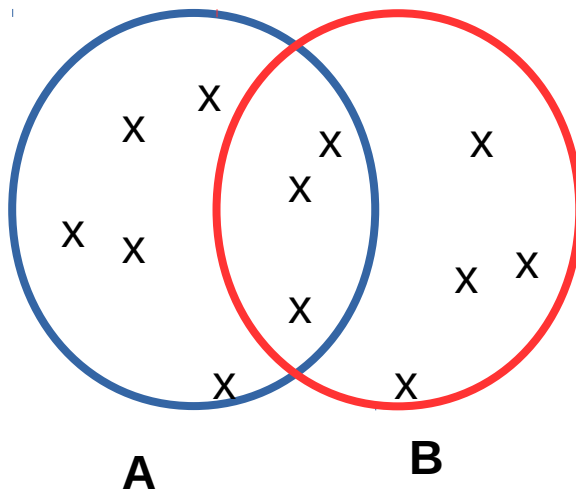
$J(A,B)$ measures:

- Similarity of data sets
- Similarity of asymmetric binary attributes (vector)
- Similarity of one value attributes (vector)
- Similarity of categorical attributes with dummy variable transformation

Jaccard Distance : $d_j(A,B) = 1 - J(A,B)$



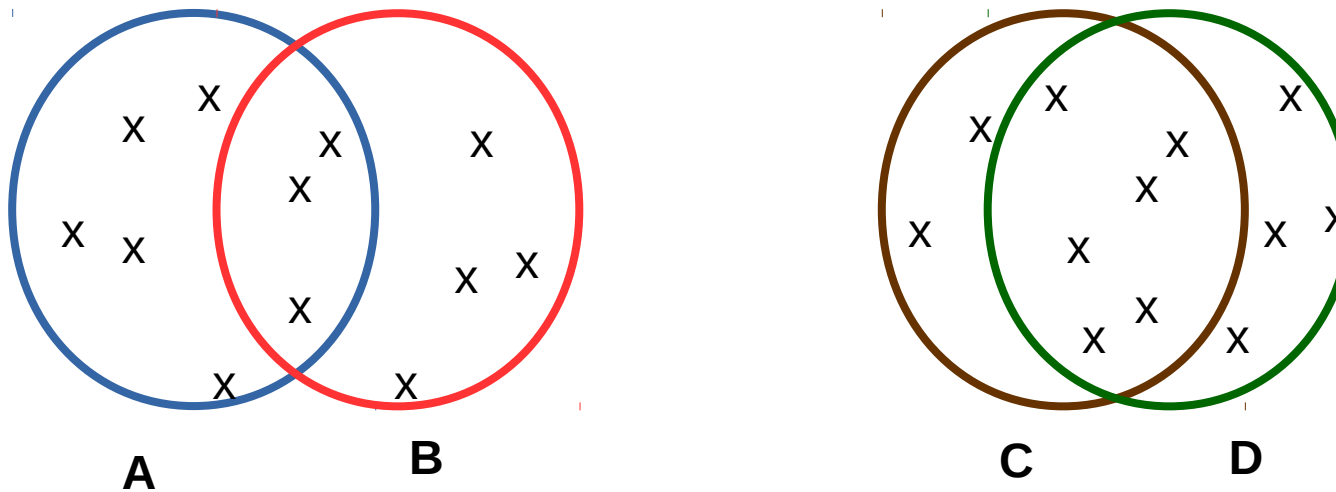
Jaccard Similarity Coefficient **- Set**



Similarity of data sets



Jaccard Similarity Coefficient - Set



$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}.$$

Similarity of data sets :

$$J(A, B) = 3/12 = 0,25$$

$$J(C, D) = 6/12 = 0,5$$



Jaccard Similarity Coefficient - Set

$$A = \{1, 2, 5, 8, 9, 10\},$$
$$B = \{3, 5, 7, 8, 12, 13\}$$

$$A \cap B = \{5, 8\}$$

$$J(A, B) = |A \cap B| / (|A| + |B| - |A \cap B|)$$
$$= 2 / (6 + 6 - 2) = 0,2$$



Jaccard Similarity Coefficient

- *asymmetric* binary vector

https://en.wikipedia.org/wiki/Jaccard_index

M_{11} represents the total number of attributes where A and B both have a value of 1.

M_{01} represents the total number of attributes where the attribute of A is 0 and the attribute of B is 1.

M_{10} represents the total number of attributes where the attribute of A is 1 and the attribute of B is 0.

M_{00} represents the total number of attributes where A and B both have a value of 0.

Each attribute must fall into one of these four categories, meaning that

$$M_{11} + M_{01} + M_{10} + M_{00} = n.$$

The Jaccard similarity coefficient, J , is given as

$$J = \frac{M_{11}}{M_{01} + M_{10} + M_{11}}.$$

The Jaccard distance, d_J , is given as

$$d_J = \frac{M_{01} + M_{10}}{M_{01} + M_{10} + M_{11}} = 1 - J.$$

		A	
		0	1
B	0	M_{00}	M_{10}
	1	M_{01}	M_{11}



Jaccard Similarity Coefficient

- asymmetric binary vector

	Fever	S1	S2	S3	Test1	Test2	Test3
Patient1	y	y	n	y	p	p	n
Patient2	y	n	y	y	n	p	p



	Fever	S1	S2	S3	Test1	Test2	Test3
Patient1	1	1	0	1	1	1	0
Patient2	1	0	1	1	0	1	1

$$J(\text{Patient1}, \text{Patient2}) = (1+1+1)/(3+2+2) = 0,43$$



Simple Matching Coefficient (SMC)

$$\begin{aligned} \text{SMC} &= \frac{\text{number of matching attributes}}{\text{number of attributes}} \\ &= \frac{M_{00} + M_{11}}{M_{00} + M_{01} + M_{10} + M_{11}} \end{aligned}$$

Measure the similarity of symmetric binary attributes

Biased with large M_{00} against asymmetric binary attributes



Jaccard Similarity Coefficient - Generalization (Weighted)

$$J_w(\mathbf{x}, \mathbf{y}) = \frac{\sum_i \min(x_i, y_i)}{\sum_i \max(x_i, y_i)},$$

	Test1 (mg)	S1	S2	S3
Patient1	1	1	0	1
Patient2	2	0	1	1

$$J_w(\text{Patient1}, \text{Patient2}) = (1+0+0+1)/(2+1+1+1)=2/5=0,4$$



Cosine Similarity

$$\mathbf{A} \cdot \mathbf{B} = \|\mathbf{A}\| \|\mathbf{B}\| \cos \theta$$

$$\text{similarity} = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}},$$

$$\text{angular distance} = \frac{\cos^{-1}(\text{cosine similarity})}{\pi}$$

If the vector elements are always positive:

$$\text{angular distance} = \frac{2 \cdot \cos^{-1}(\text{cosine similarity})}{\pi}$$



Cosine Similarity

Example

(Paris, Lyon) in documents :

Doc1 (3, 2), Doc2 (2, 3),

	Doc1	Doc2	Degres
Paris	3	2	22,62
Lyon	2	3	CosineD
			0,92



Cosine Similarity

Example

(Paris, Lyon) in documents :

Doc2 (2, 3), Doc3 (4, 6)

	Doc3	Doc2	Degres
Paris	4	2	0,00
Lyon	6	3	CosineD
			1,00



Cosine Similarity

Example

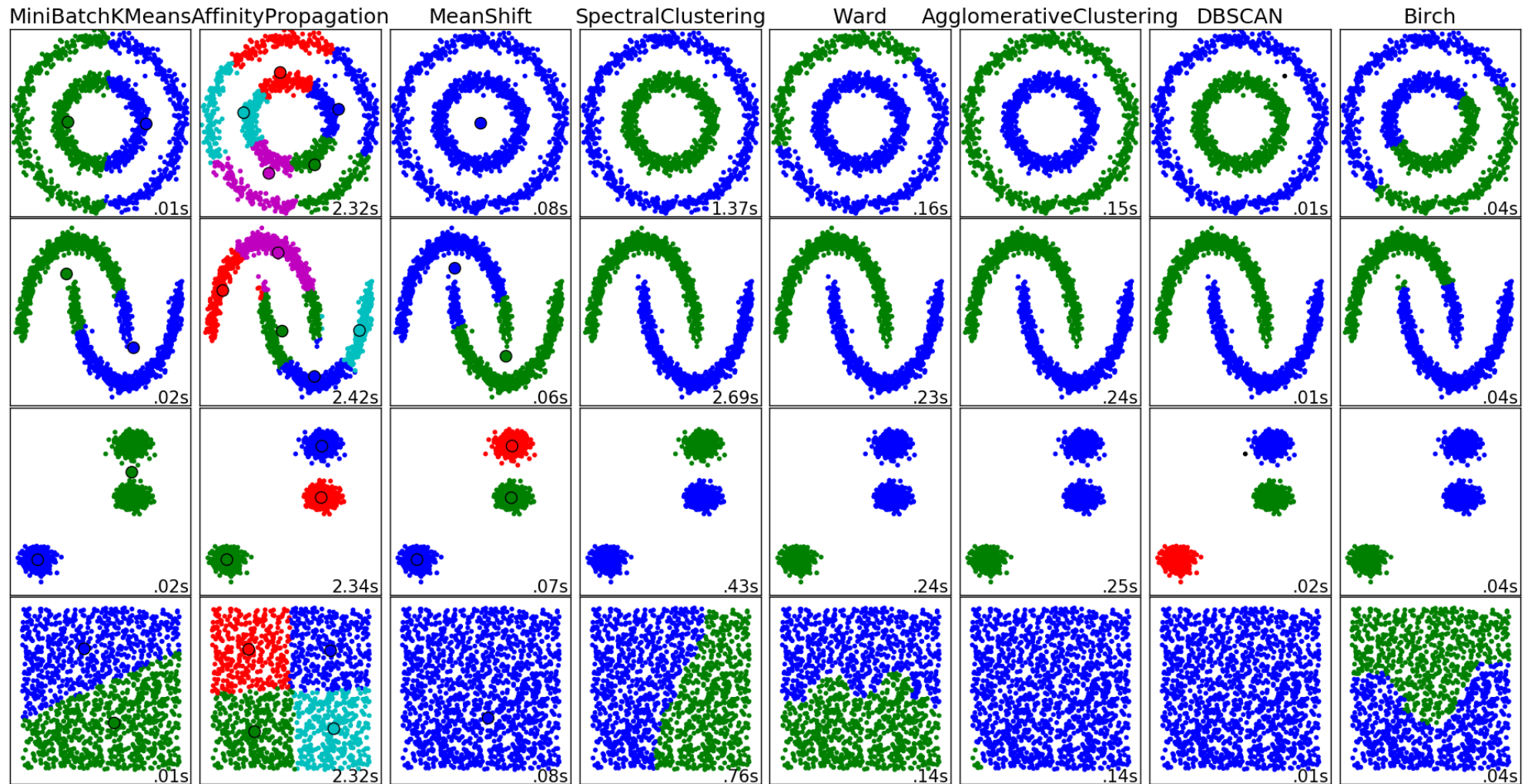
(Paris, Lyon) in documents :

Doc4 (2, 0), Doc5 (0, 5)

	Doc4	Doc5	Degres
Paris	2	0	90,00
Lyon	0	5	CosineD
			0,00



Clustering



Clustering Models

Centroid models :

K-means, k-medoids ...

Connectivity models :

Hierarchical clustering...

Density models :

DBSCAN, OPTICS ...

Graph-based models :

HCS clustering ...

....



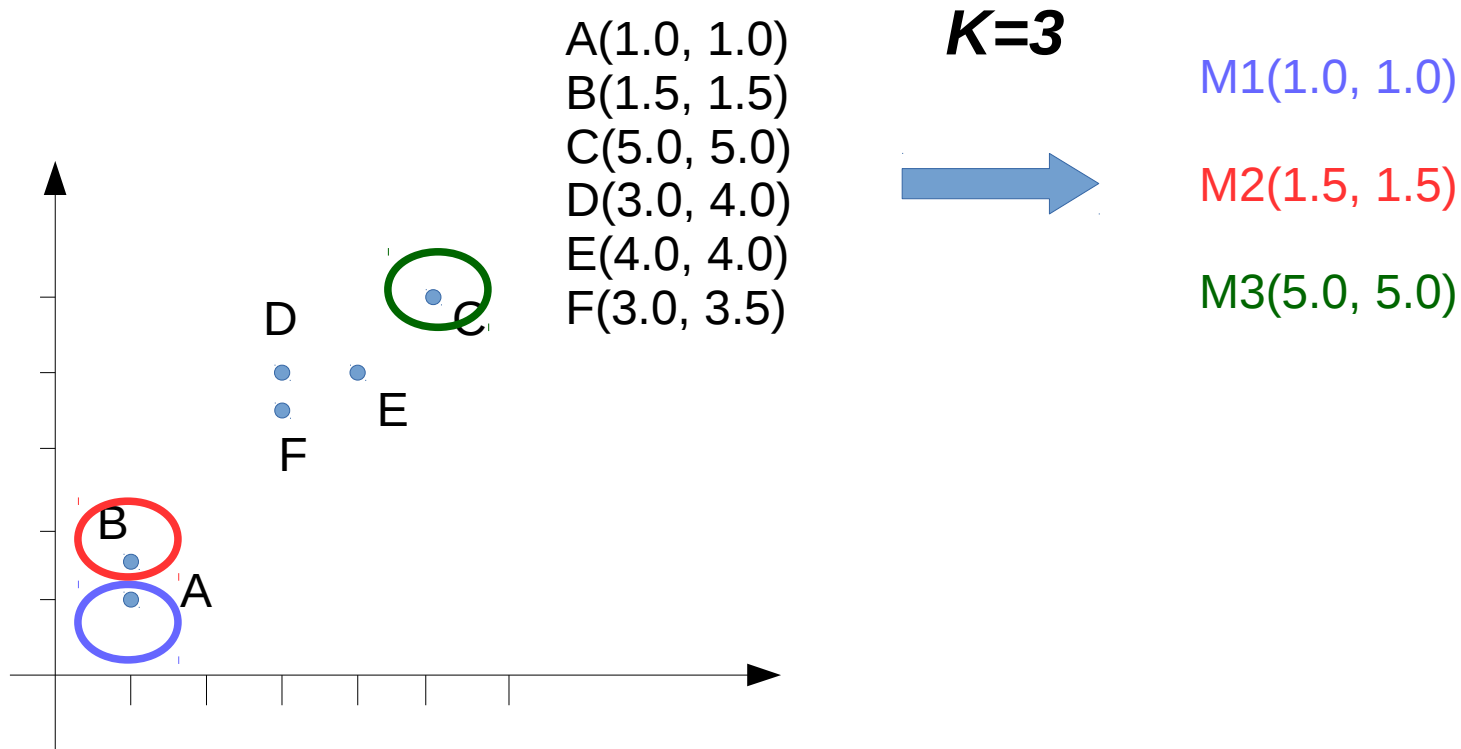
K-Means

Given a positive integer K , the *k-means* algorithm works with the following steps :

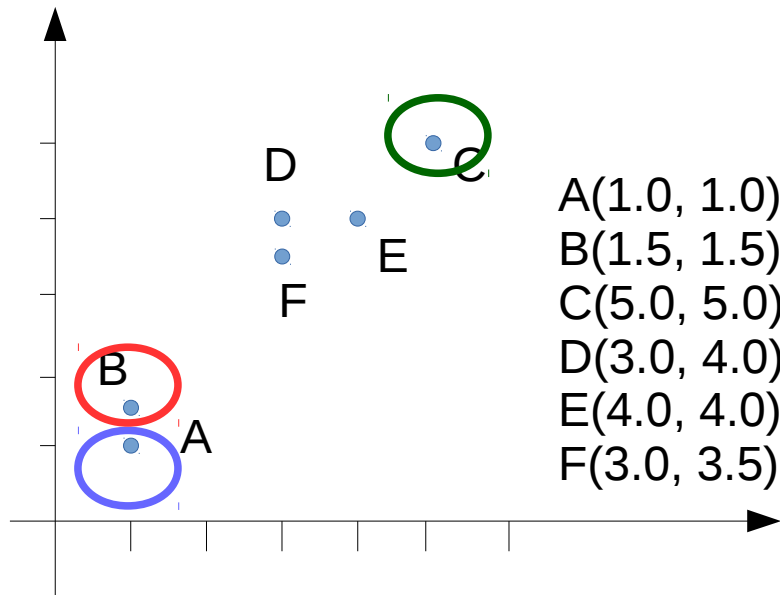
1. Initialize k points randomly as the means for the beginning.
2. Assign each data point to its nearest mean to form a cluster.
3. For each cluster, compute its new means
4. Go back to Step2 till that the means don't change



K-Means Example (1)



K-Means Example (2)



K=3

M1(1.0, 1.0)

M2(1.5, 1.5)

M3(5.0, 5.0)

$$d_M(M1, D) = 5.0$$

$$d_M(M1, E) = 6.0$$

$$d_M(M1, F) = 4.5$$

$$d_M(M2, D) = 4.0$$

$$d_M(M2, E) = 5.0$$

$$d_M(M2, F) = 3.5$$

$$d_M(M3, D) = 3.0$$

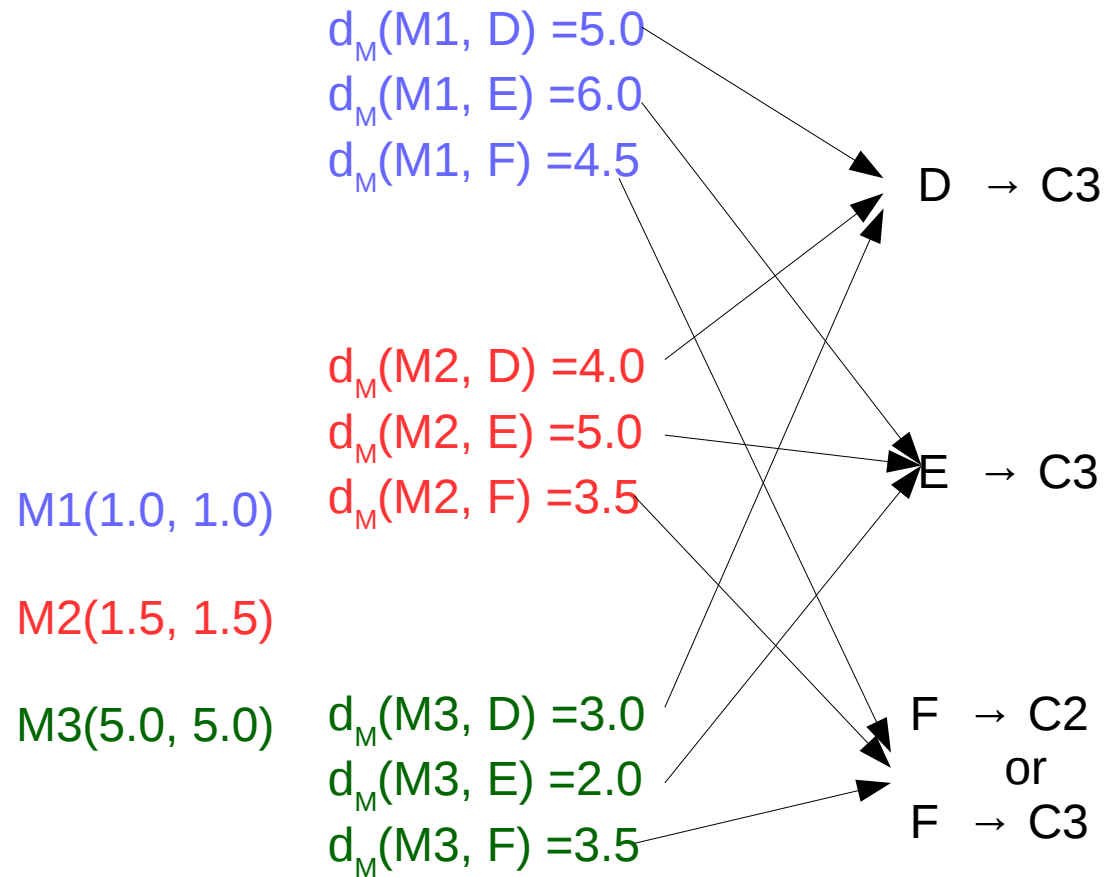
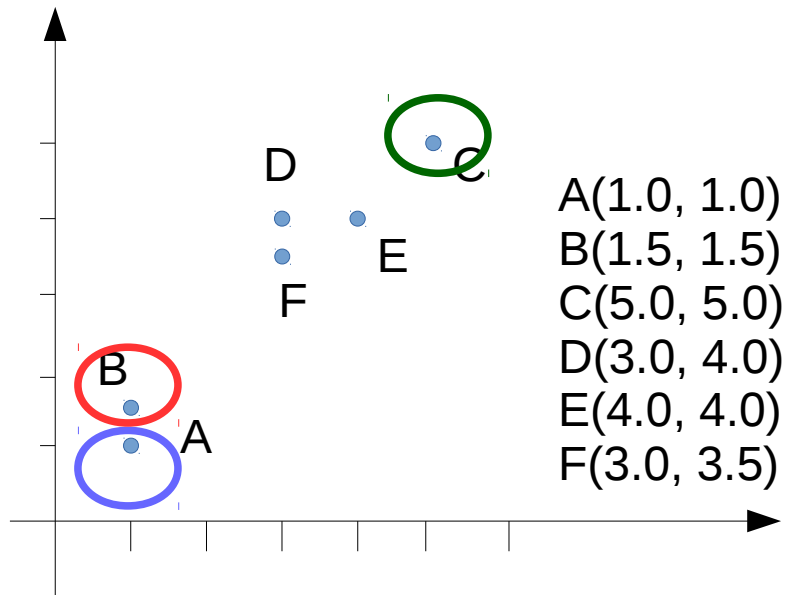
$$d_M(M3, E) = 2.0$$

$$d_M(M3, F) = 3.5$$

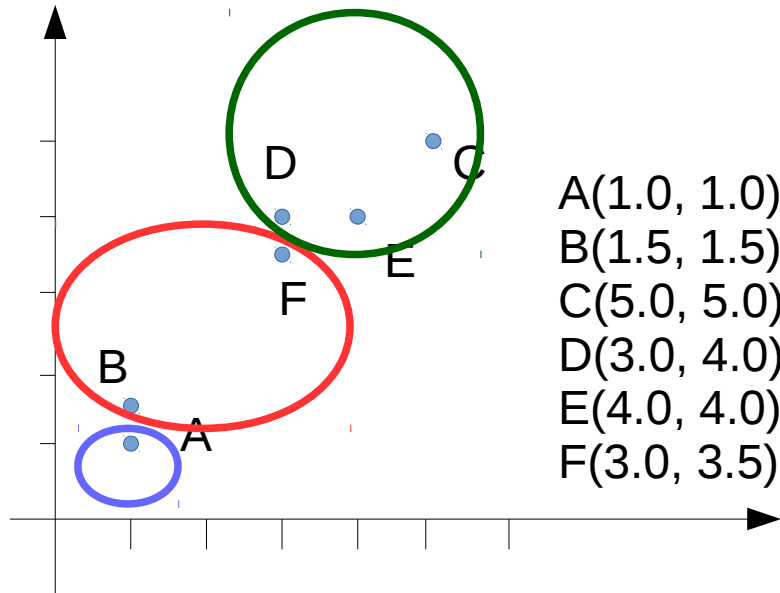
Manhattan Distance



K-Means Example (3)



K-Means Example (4)



C1 : A

C2 : B

F

C3 : C

D

E



M1(1.0, 1.0)

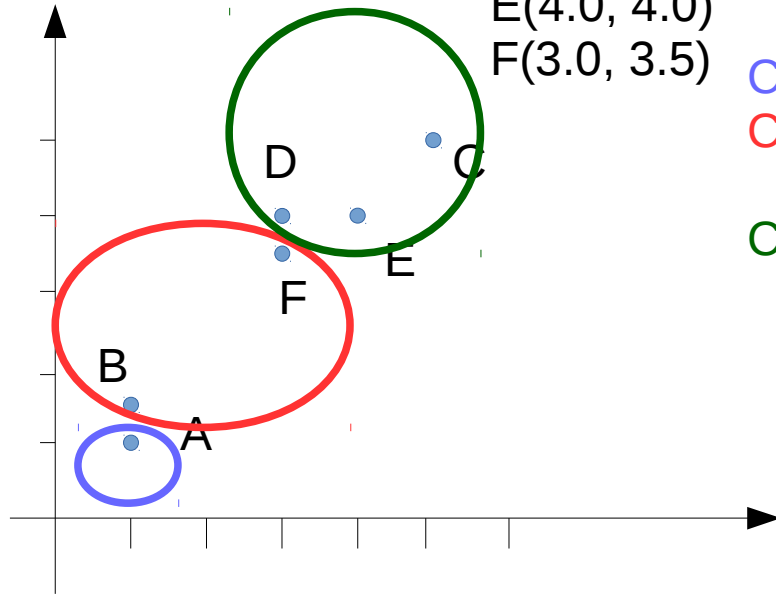
M2(2.25, 2.5)

M3(4.0, 4.33)



K-Means Example (5)

A(1.0, 1.0)
B(1.5, 1.5)
C(5.0, 5.0)
D(3.0, 4.0)
E(4.0, 4.0)
F(3.0, 3.5)



C1 : A
C2 : B
F
C3 : C
D
E

M1(1.0, 1.0)

M2(2.25, 2.5)

M3(4.0, 4.33)

$$d_M(M1, A) = 0.0$$

$$d_M(M1, B) = 1.0$$

$$d_M(M1, C) = 8.0$$

$$d_M(M1, D) = 5.0$$

$$d_M(M1, E) = 6.0$$

$$d_M(M1, F) = 4.5$$

$$d_M(M2, A) = 2.75$$

$$d_M(M2, B) = 1.75$$

$$d_M(M2, C) = 5.25$$

$$d_M(M2, D) = 2.25$$

$$d_M(M2, E) = 3.25$$

$$d_M(M2, F) = 1.75$$

$$d_M(M3, A) = 6.33$$

$$d_M(M3, B) = 5.33$$

$$d_M(M3, C) = 1.67$$

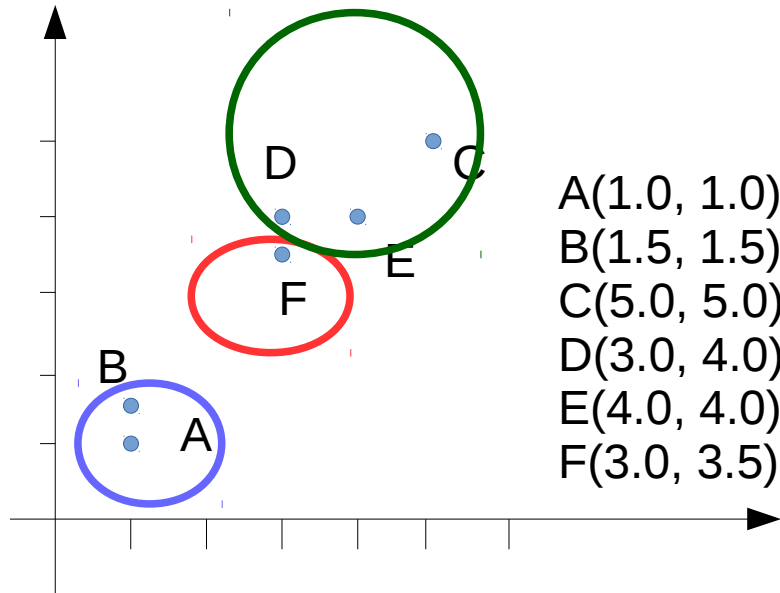
$$d_M(M3, D) = 1.33$$

$$d_M(M3, E) = 0.33$$

$$d_M(M3, F) = 1.83$$



K-Means Example (6)



C1 : A
B
C2 : F
C3 : C
D
E



M1(1.25, 1.25)

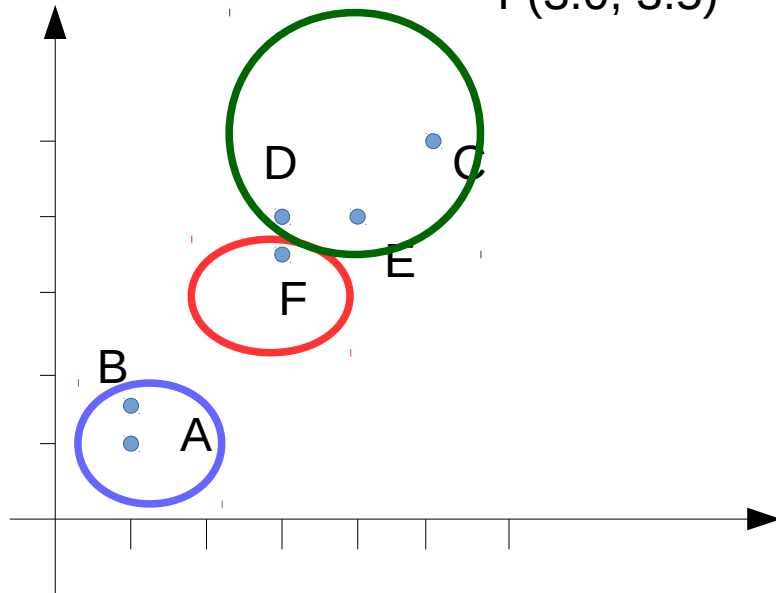
M2(3.0, 3.5)

M3(4.0, 4.33)



K-Means Example (7)

A(1.0, 1.0)
B(1.5, 1.5)
C(5.0, 5.0)
D(3.0, 4.0)
E(4.0, 4.0)
F(3.0, 3.5)



$d_M(M1, A) = 0.5$
 $d_M(M1, B) = 0.5$
 $d_M(M1, C) = 7.5$
 $d_M(M1, D) = 4.5$
 $d_M(M1, E) = 5.5$
 $d_M(M1, F) = 4.0$

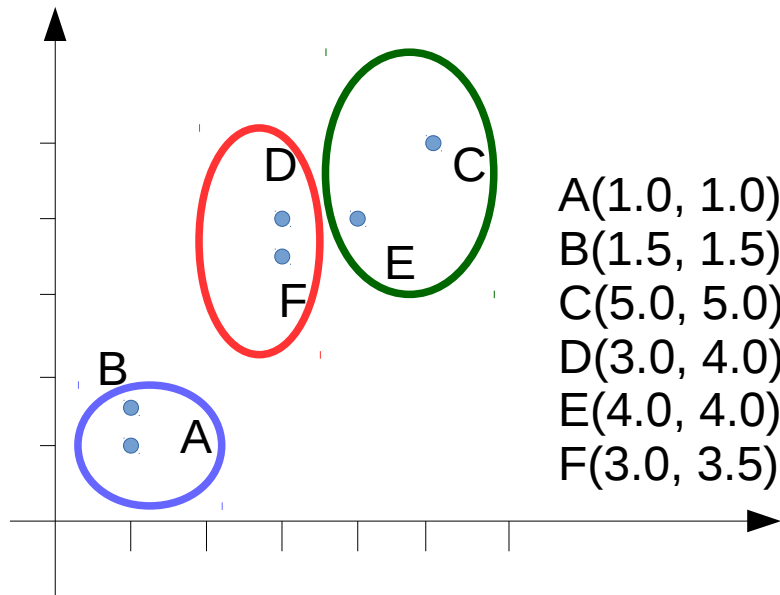
C1 : A M1(1.25, 1.25)
 B
C2 : F M2(3.0, 3.5)
C3 : C
 D M3(4.0, 4.33)
 E

$d_M(M2, A) = 4.5$
 $d_M(M2, B) = 3.5$
 $d_M(M2, C) = 3.5$
 $d_M(M2, D) = 0.5$
 $d_M(M2, E) = 1.5$
 $d_M(M2, F) = 0$

$d_M(M3, A) = 6.33$
 $d_M(M3, B) = 5.33$
 $d_M(M3, C) = 1.67$
 $d_M(M3, D) = 1.33$
 $d_M(M3, E) = 0.33$
 $d_M(M3, F) = 1.83$



K-Means Example (8)



C1 : A
B

C2 : D
F

C3 : C
E



M1(1.25, 1.25)

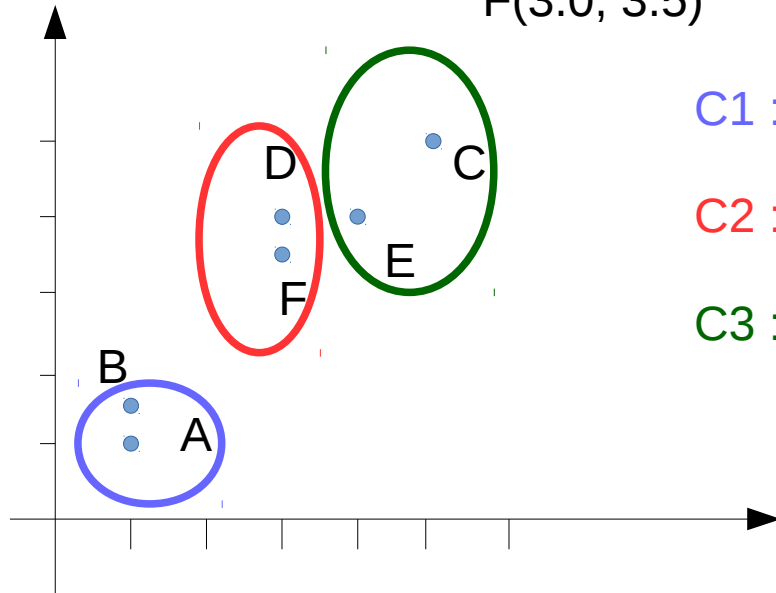
M2(3.0, 3.75)

M3(4.5, 4.5)



K-Means Example (9)

A(1.0, 1.0)
B(1.5, 1.5)
C(5.0, 5.0)
D(3.0, 4.0)
E(4.0, 4.0)
F(3.0, 3.5)



C1 : A
B
C2 : D
F
E
C3 : C
E

M1(1.25, 1.25)

M2(3.0, 3.75)

M3(4.5, 4.5)

$$d_M(M1, A) = 0.5$$

$$d_M(M1, B) = 0.5$$

$$d_M(M1, C) = 7.5$$

$$d_M(M1, D) = 4.5$$

$$d_M(M1, E) = 5.5$$

$$d_M(M1, F) = 4.0$$

$$d_M(M2, A) = 4.75$$

$$d_M(M2, B) = 3.75$$

$$d_M(M2, C) = 3.25$$

$$d_M(M2, D) = 0.25$$

$$d_M(M2, E) = 1.25$$

$$d_M(M2, F) = 0.25$$

$$d_M(M3, A) = 7$$

$$d_M(M3, B) = 6$$

$$d_M(M3, C) = 1$$

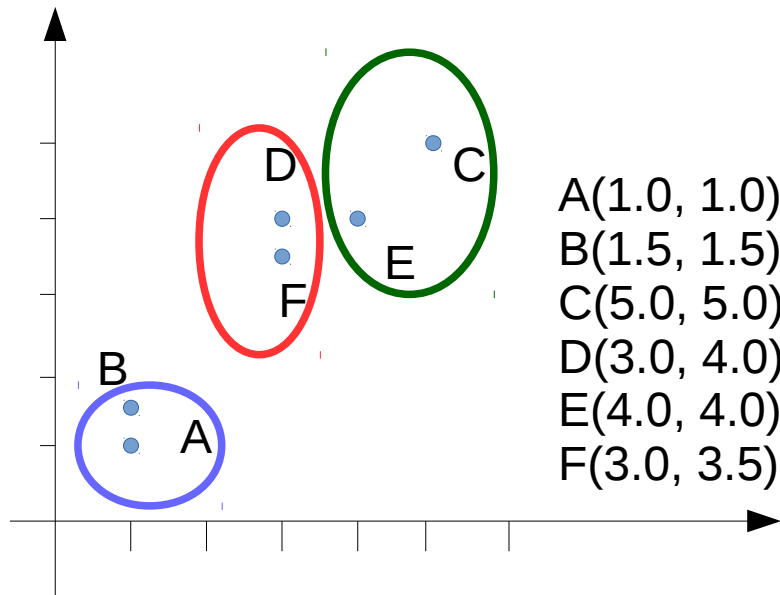
$$d_M(M3, D) = 2$$

$$d_M(M3, E) = 1$$

$$d_M(M3, F) = 2.5$$



K-Means Example (10)



C1 : A
B
C2 : D
F
C3 : C
E



M1(1.25, 1.25)

M2(3.0, 3.75)

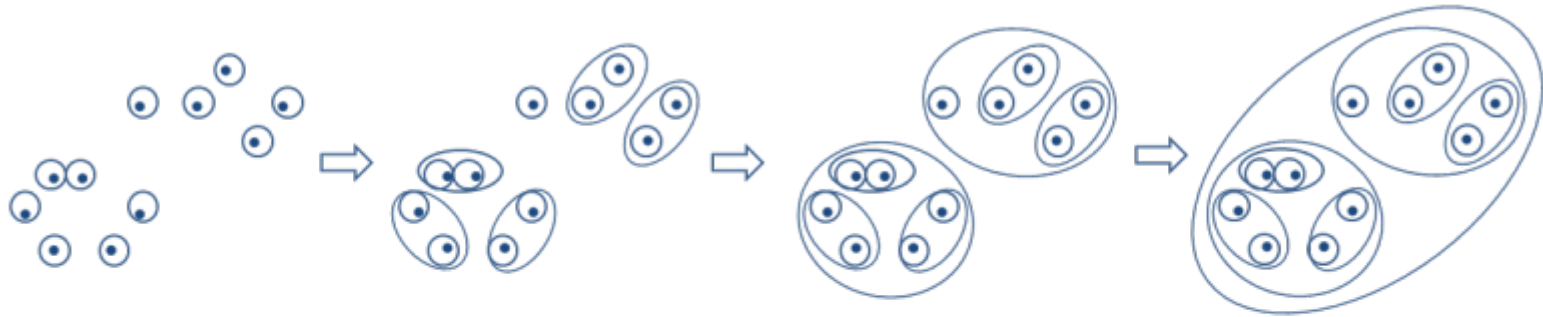
M3(4.5, 4.5)

Keep the same means!

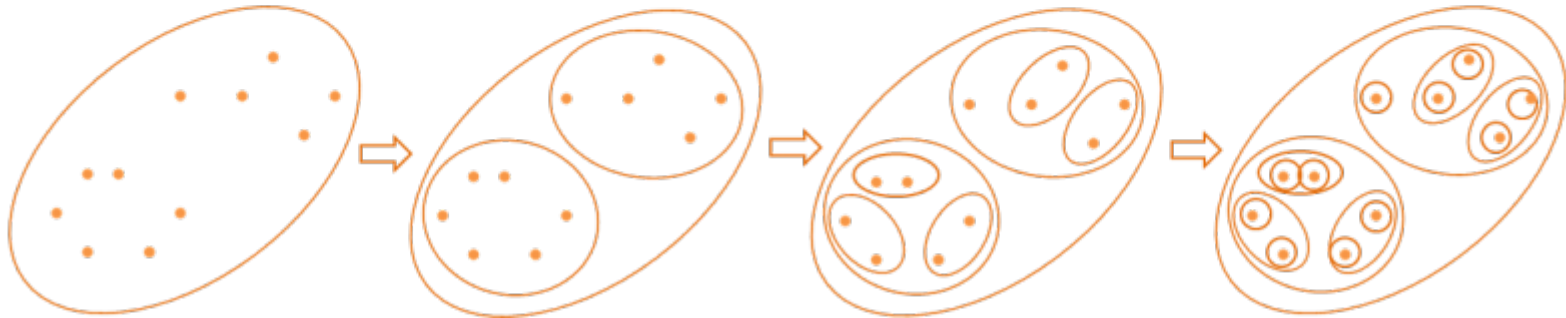


Hierarchical Clustering

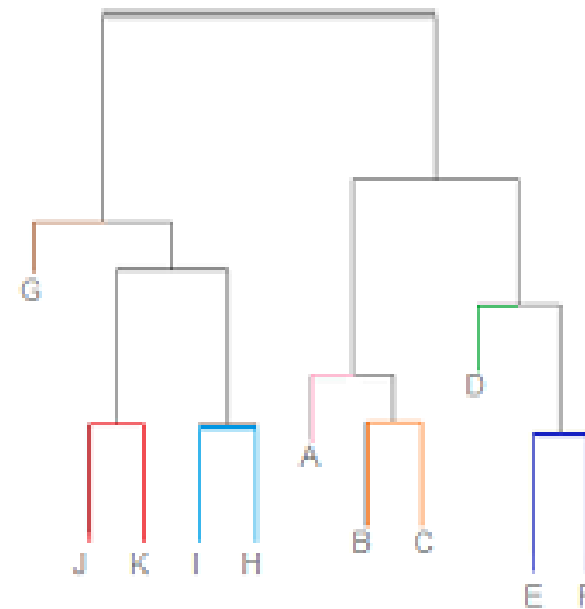
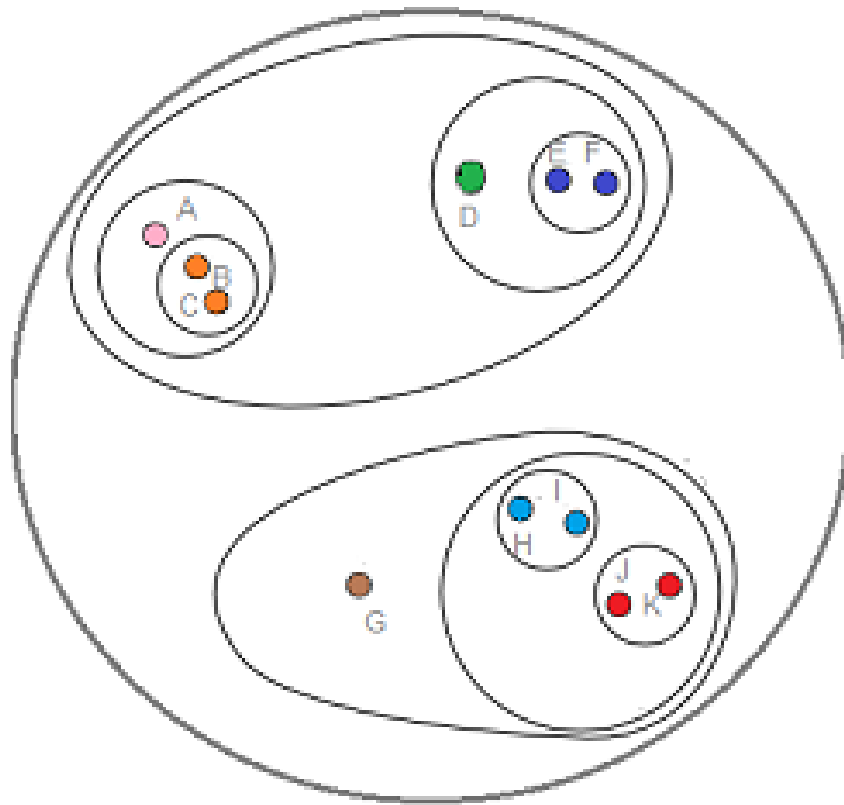
Agglomerative Hierarchical Clustering



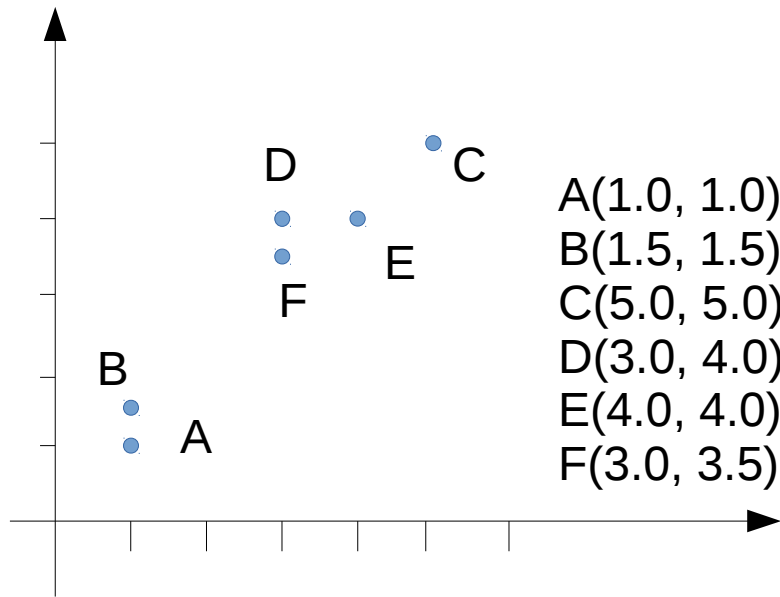
Divisive Hierarchical Clustering



Agglomerative Clustering



Agglomerative Hierarchical Clustering (1)



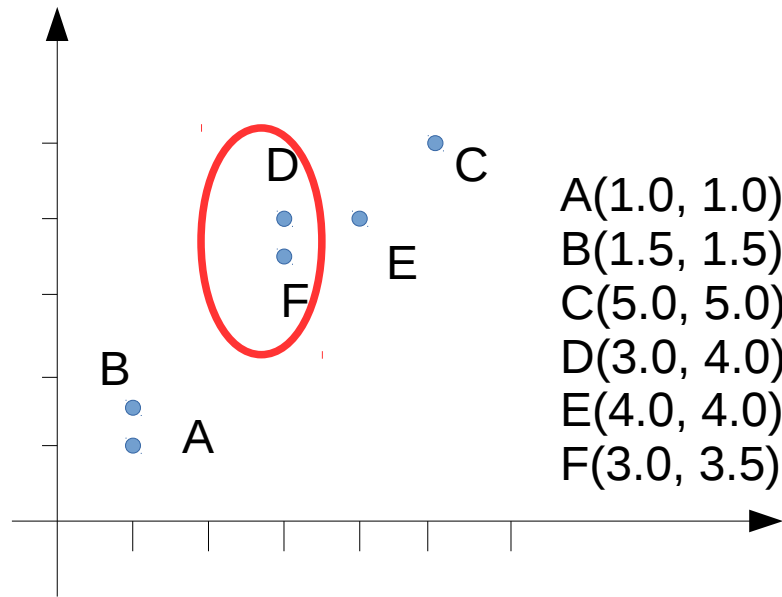
Manhattan Distance Matrix

	A	B	C	D	E
A	0				
B	1	0			
C	8	7	0		
D	5	4	3	0	
E	6	5	2	1	0
F	4.5	3.5	3.5	0.5	1.5

Goal : 3 Clusters



Agglomerative Hierarchical Clustering (2)



Manhattan Distance Matrix

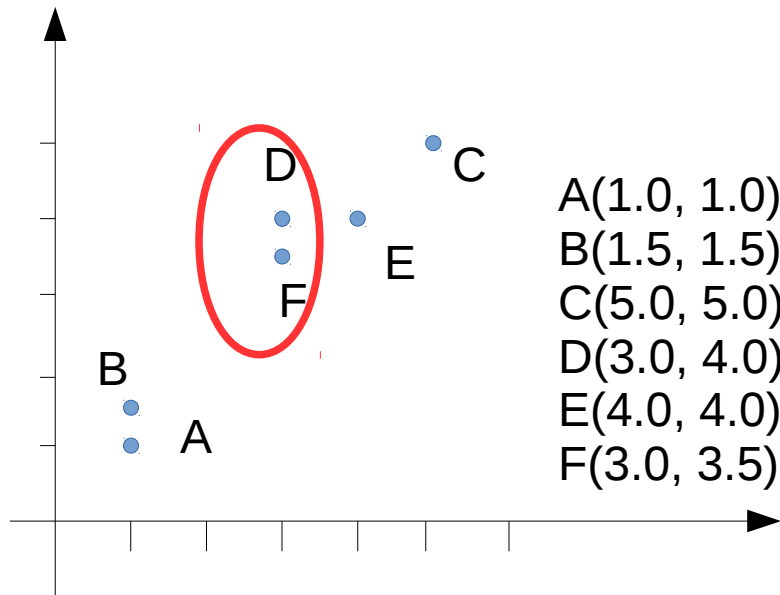
	A	B	C	D	E
A	0				
B	1	0			
C	8	7	0		
D	5	4	3	0	
E	6	5	2	1	0
F	4.5	3.5	3.5	0.5	1.5

Min

Goal : 3 Clusters



Agglomerative Hierarchical Clustering (3)



Goal : 3 Clusters

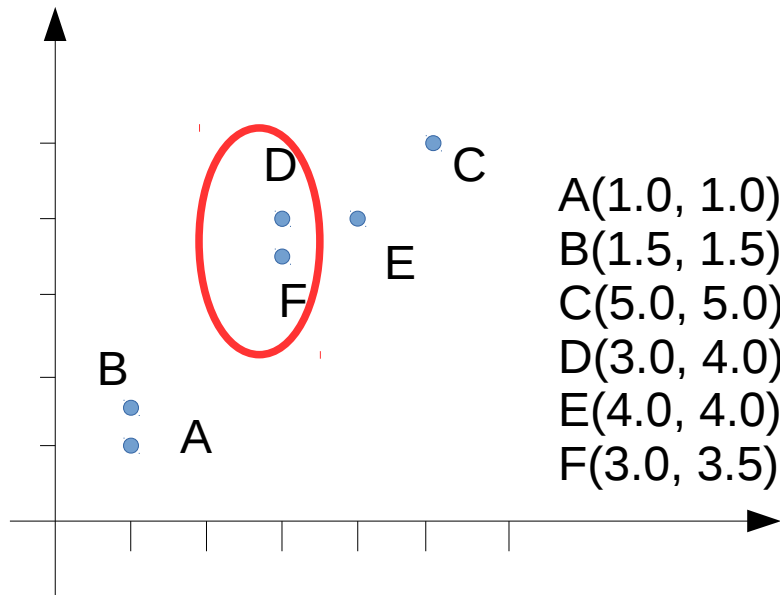
Manhattan Distance Matrix

	A	B	C	DF
A	0			
B	1	0		
C	8	7	0	
DF	4.5	3.5	3	
E	6	5	2	1

Minimum measure for group distance



Agglomerative Hierarchical Clustering (4)



Goal : 3 Clusters

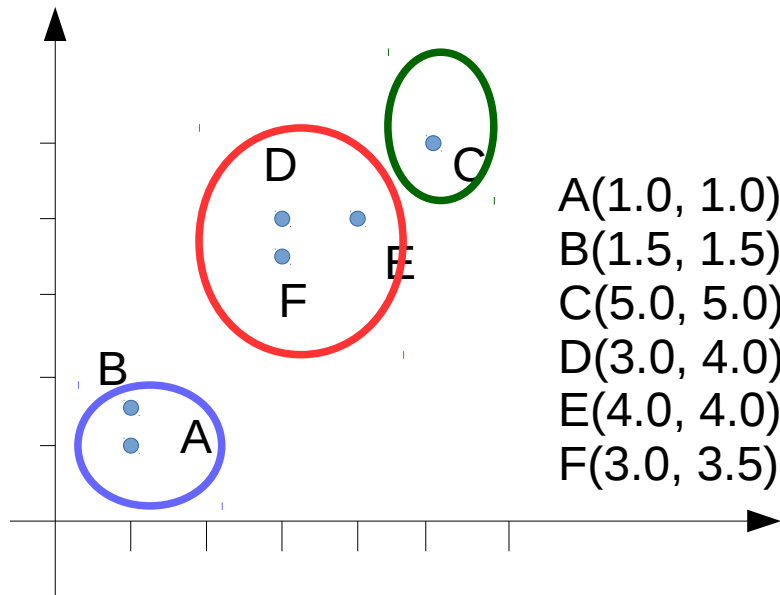
Manhattan Distance Matrix

	A	B	C	DF
A	0			
B	1	0		
C	8	7	0	
DF	4.5	3.5	3	
E	6	5	2	1

Min



Agglomerative Hierarchical Clustering (5)



Goal : 3 Clusters

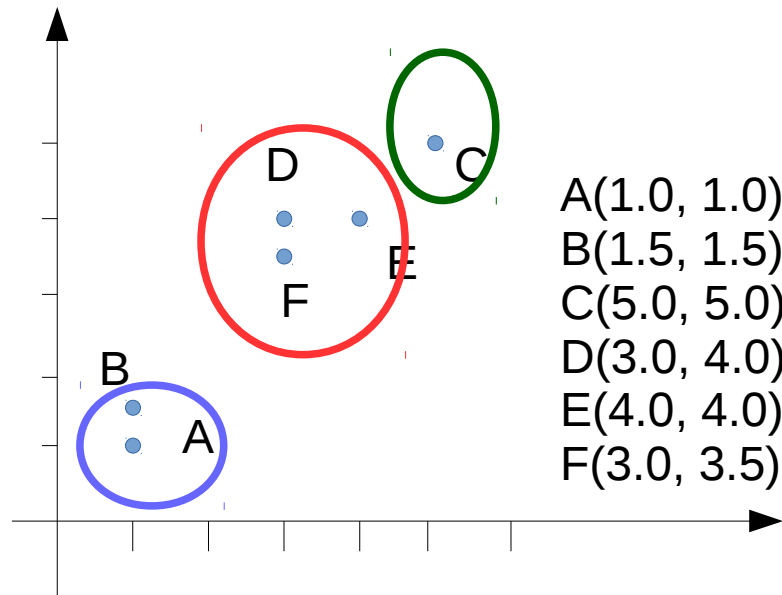
Manhattan Distance Matrix

	A	B	C	DF
A	0			
B	1	0		
C	8	7	0	
DF	4.5	3.5	3	
E	6	5	2	1

Min



Agglomerative Hierarchical Clustering (6)



Manhattan Distance Matrix

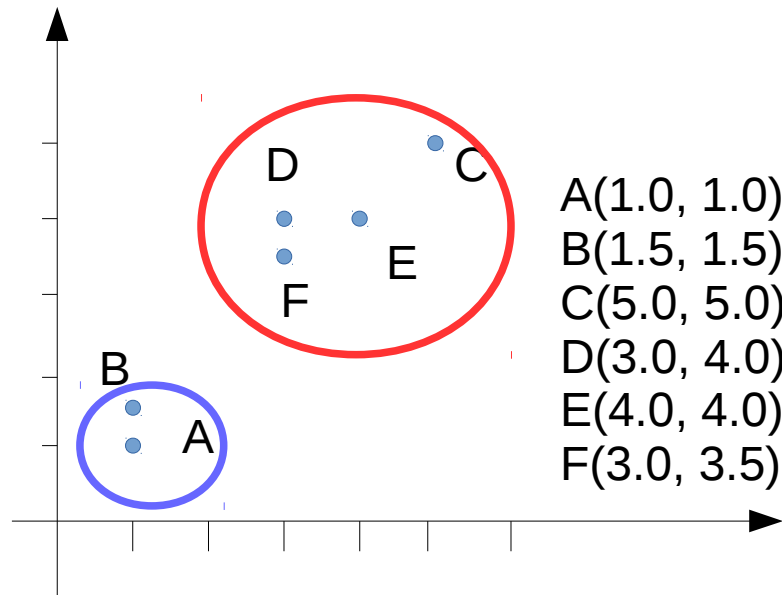
	AB	C
AB	0	7
C	7	0
DEF	3.5	2

Min

Continue the procedure ...



Agglomerative Hierarchical Clustering (7)



Manhattan Distance Matrix

	AB	C
AB	0	
C	7	0
DEF	3.5	2

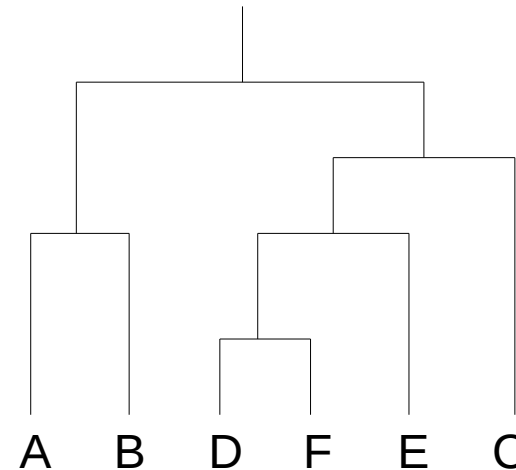
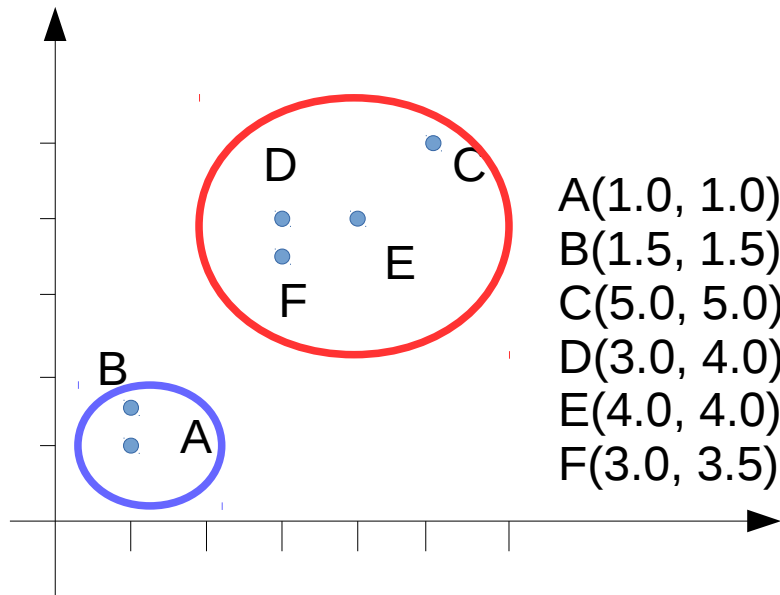
Min

Continue the procedure ...



Agglomerative Hierarchical Clustering (8)

Clustering Dendrogram

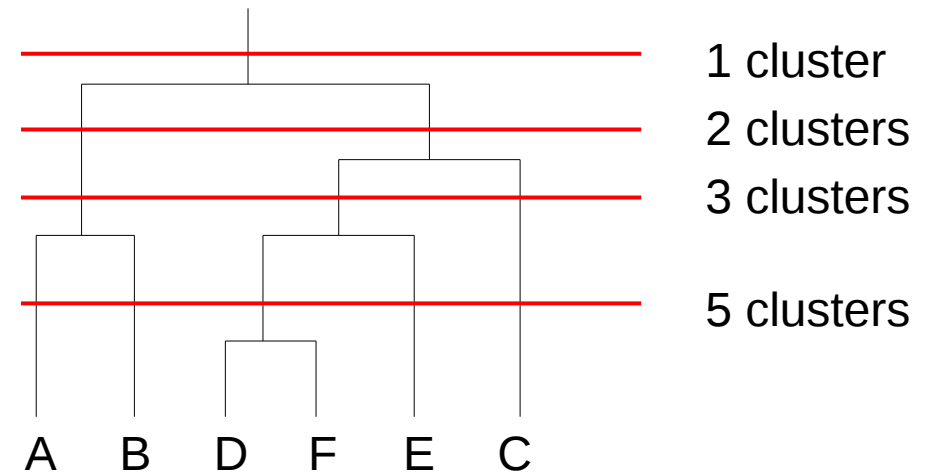
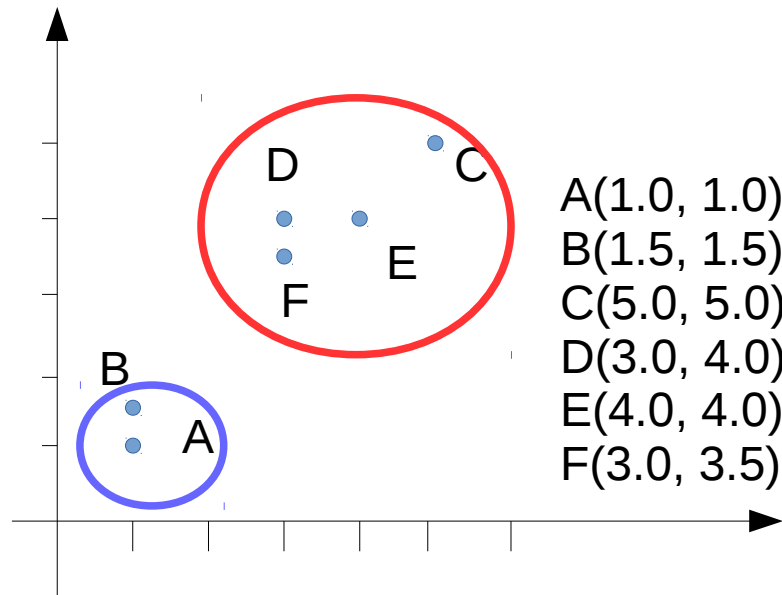


Continue the procedure ...



Agglomerative Hierarchical Clustering (9)

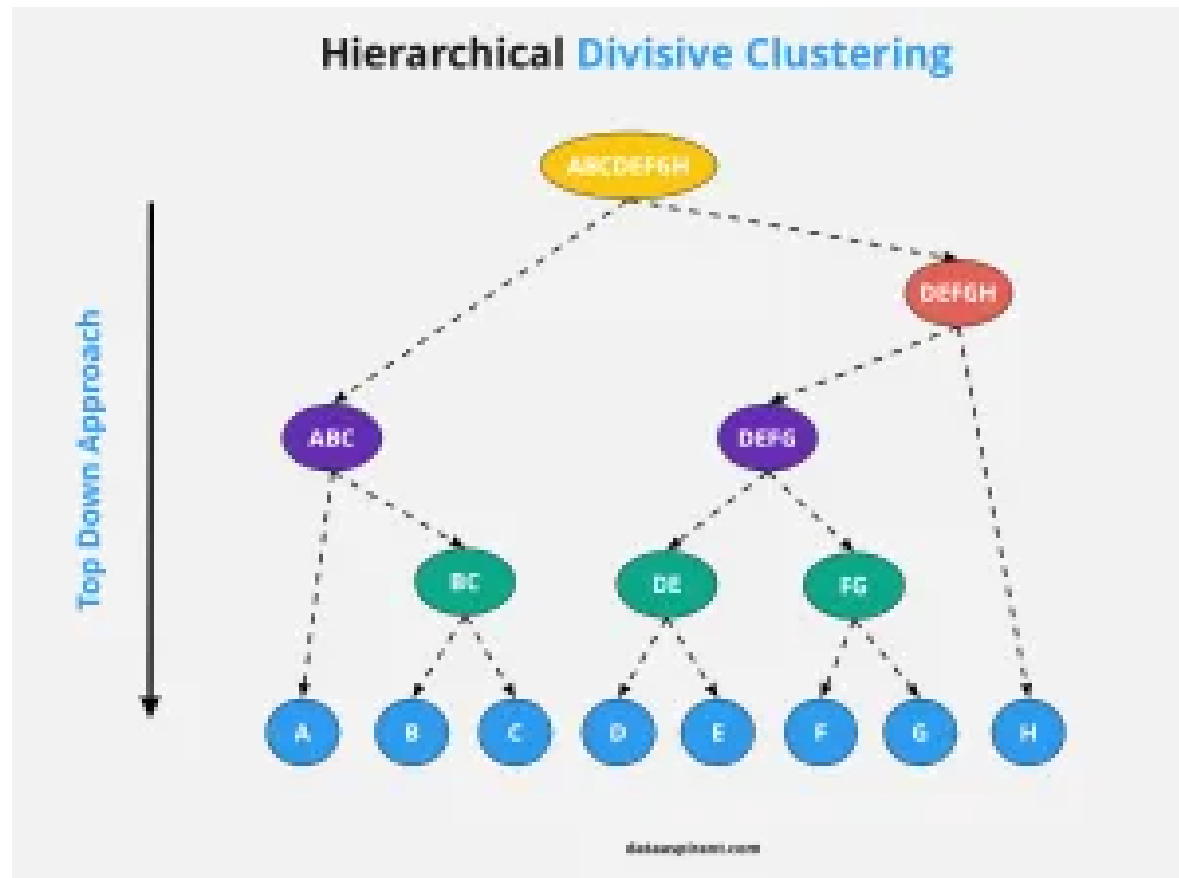
Clustering Dendrogram



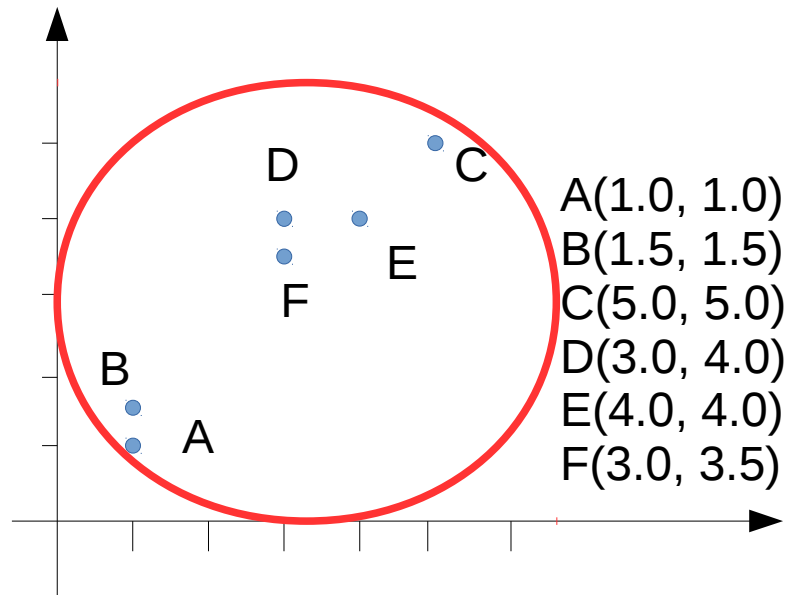
Continue the procedure ...



Hierarchical Divisive Clustering



Divisive Hierarchical Clustering (1)



Goal : 3 Clusters

Manhattan Distance Matrix

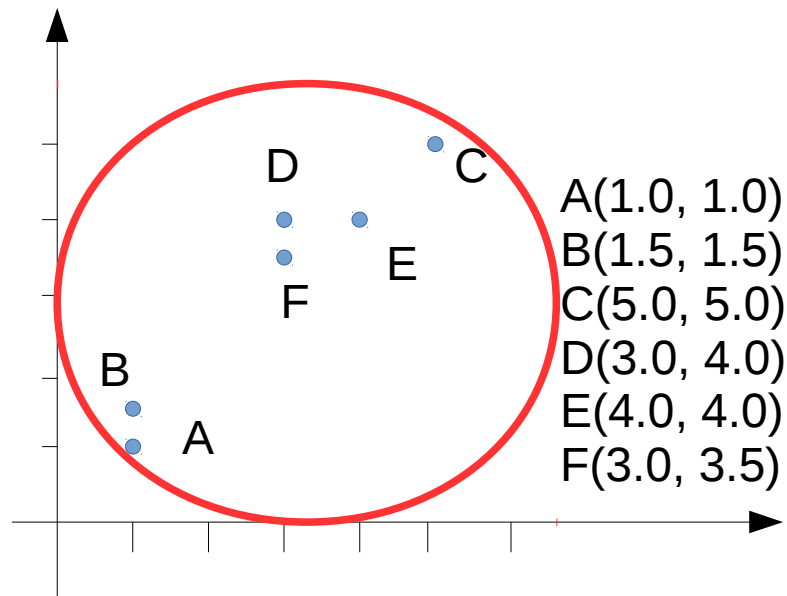
	A	B	C	D	E
A	0				
B	1	0			
C	8	7	0		
D	5	4	3		
E	6	5	2	1	
F	4.5	3.5	3.5	0.5	1.5

Manhattan Distance Average

A	B	C	D	E	F
4.9	4.1	4.7	2.7	3.1	2.7



Divisive Hierarchical Clustering (2)



Goal : 3 Clusters

Manhattan Distance Matrix

	A	B	C	D	E
A	0				
B	1	0			
C	8	7	0		
D	5	4	3	0	
E	6	5	2	1	0
F	4.5	3.5	3.5	0.5	1.5

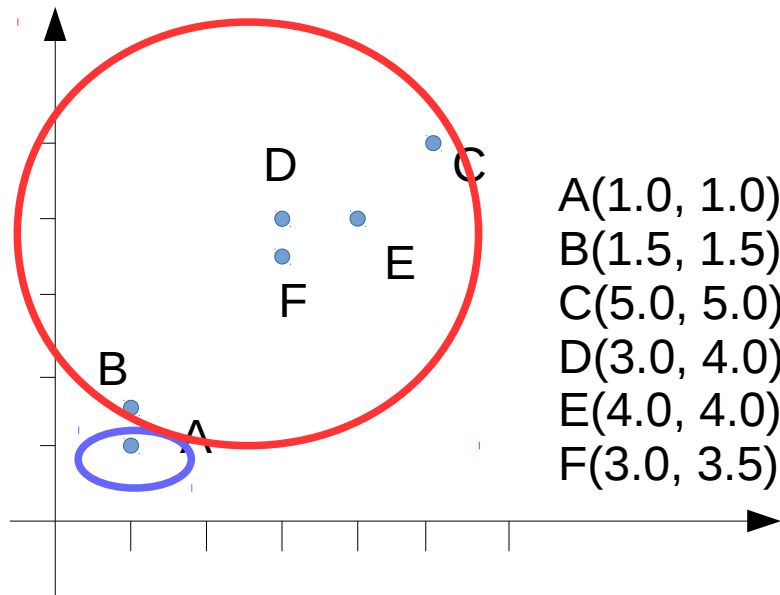
Manhattan Distance Average

A	B	C	D	E	F
4.9	4.1	4.7	2.7	3.1	2.7

Max



Divisive Hierarchical Clustering (3)



Goal : 3 Clusters

Manhattan Distance Matrix

	A	B	C	D	E
A	0				
B	1	0			
C	8	7	0		
D	5	4	3	0	
E	6	5	2	1	0
F	4.5	3.5	3.5	0.5	1.5

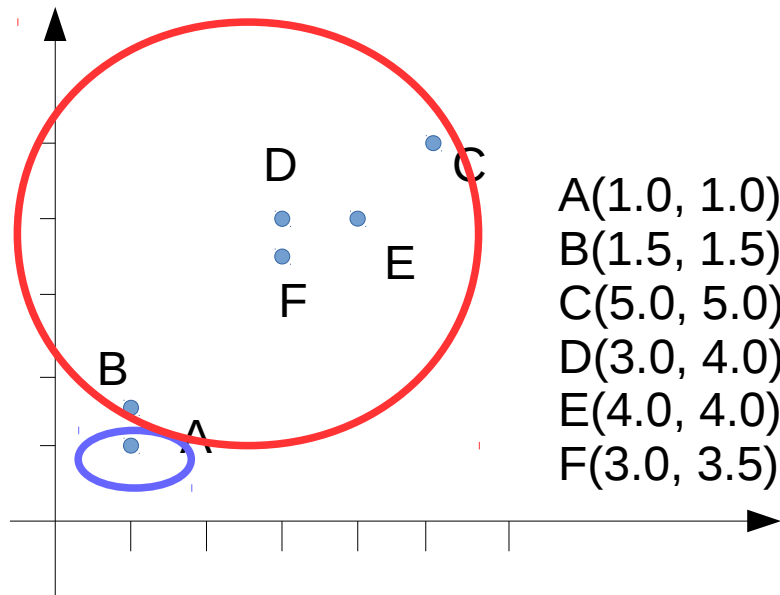
Manhattan Distance Average

A	B	C	D	E	F
4.9	4.1	4.7	2.7	3.1	2.7

Max



Divisive Hierarchical Clustering (4)



A(1.0, 1.0)
 B(1.5, 1.5)
 C(5.0, 5.0)
 D(3.0, 4.0)
 E(4.0, 4.0)
 F(3.0, 3.5)

Goal : 3 Clusters

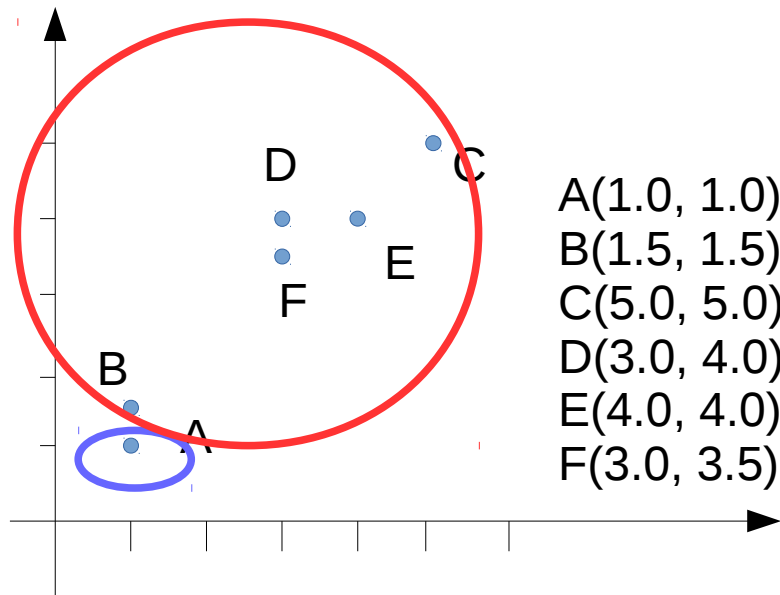
Manhattan Distance Matrix

	A	B	C	D	E
A	0				
B	1	0			
C	8	7	0		
D	5	4	3		
E	6	5	2	1	
F	4.5	3.5	3.5	0.5	1.5

	B	C	D	E	F
DistAvg	4.88	3.88	2.13	2.38	2.25
Dist(A)	1	8	5	6	4.5



Divisive Hierarchical Clustering (5)



A(1.0, 1.0)
 B(1.5, 1.5)
 C(5.0, 5.0)
 D(3.0, 4.0)
 E(4.0, 4.0)
 F(3.0, 3.5)

Goal : 3 Clusters

Manhattan Distance Matrix

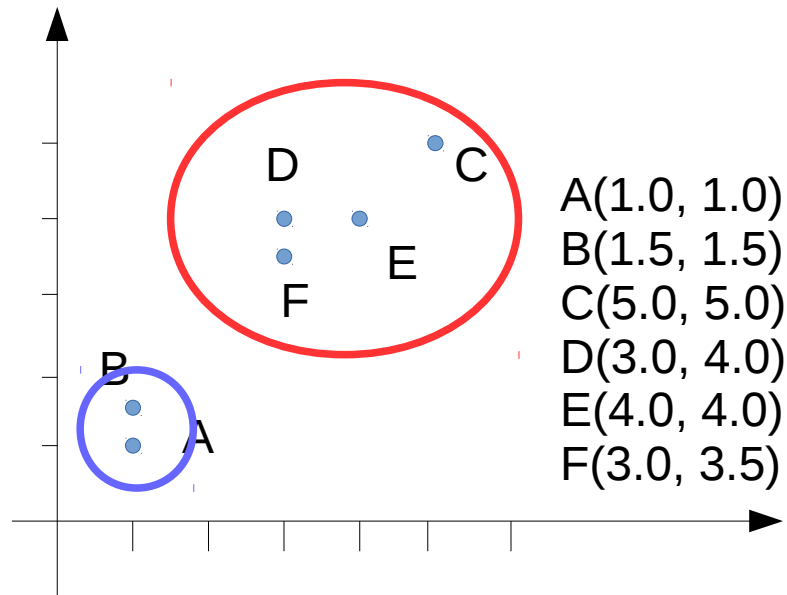
	A	B	C	D	E
A	0				
B	1	0			
C	8	7	0		
D	5	4	3	0	
E	6	5	2	1	0
F	4.5	3.5	3.5	0.5	1.5

	B	C	D	E	F
DistAvg	4.88	3.88	2.13	2.38	2.25
Dist(A)	1	8	5	6	4.5

A, B in a group



Divisive Hierarchical Clustering (6)



A(1.0, 1.0)
 B(1.5, 1.5)
 C(5.0, 5.0)
 D(3.0, 4.0)
 E(4.0, 4.0)
 F(3.0, 3.5)

Goal : 3 Clusters

Manhattan Distance Matrix

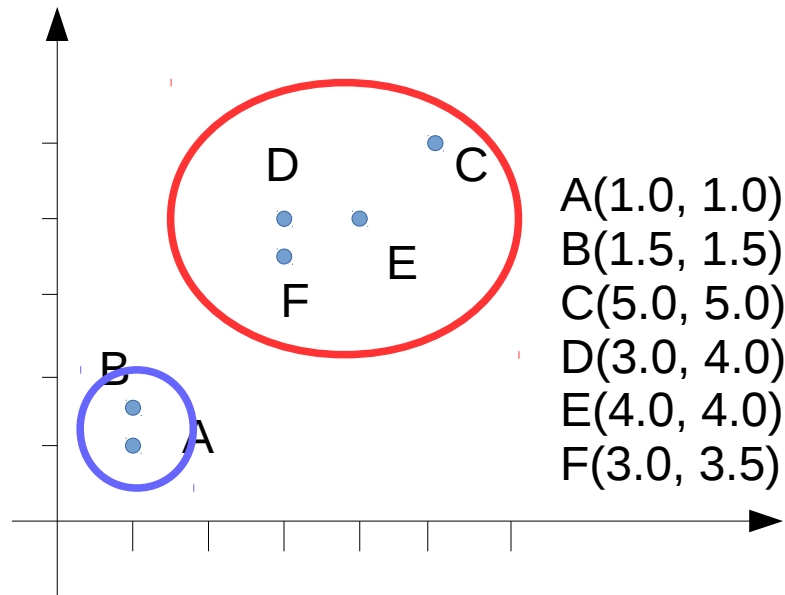
	A	B	C	D	E
A	0				
B	1	0			
C	8	7	0		
D	5	4	3		
E	6	5	2	1	
F	4.5	3.5	3.5	0.5	1.5

	B	C	D	E	F
DistAvg	4.88	3.88	2.13	2.38	2.25
Dist(A)	1	8	5	6	4.5

A, B in a group



Divisive Hierarchical Clustering (7)



Goal : 3 Clusters

Manhattan Distance Matrix

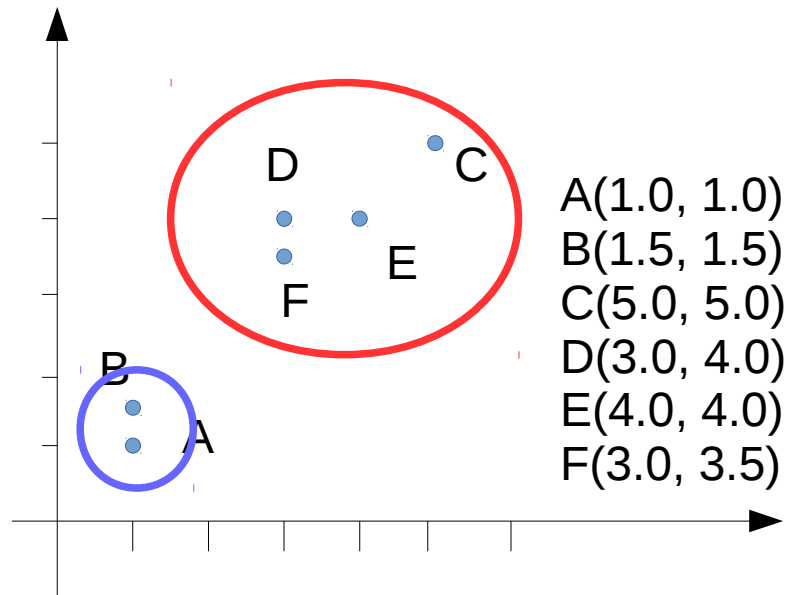
	A	B	C	D	E
A	0				
B	1	0			
C	8	7	0		
D	5	4	3		
E	6	5	2	1	
F	4.5	3.5	3.5	0.5	1.5

	C	D	E	F
DistAvg	2.83	1.5	1.5	1.83
Dist(AB)	7	4	5	3.5

C, D, E, and F will not in the group (A,B)



Divisive Hierarchical Clustering (8)



A(1.0, 1.0)
 B(1.5, 1.5)
 C(5.0, 5.0)
 D(3.0, 4.0)
 E(4.0, 4.0)
 F(3.0, 3.5)

Goal : 3 Clusters

Manhattan Distance Matrix

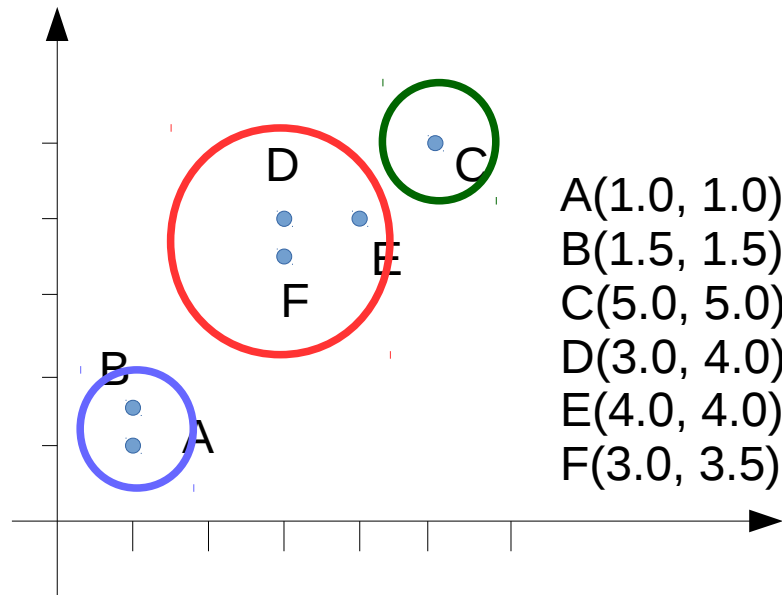
	A	B	C	D	E
A	0				
B	1	0			
C	8	7	0		
D	5	4	3		
E	6	5	2	1	
F	4.5	3.5	3.5	0.5	1.5

	C	D	E	F
DistAvg	2.83	1.5	1.5	1.83
Dist(AB)	7	4	5	3.5

Max



Divisive Hierarchical Clustering (9)



A(1.0, 1.0)
 B(1.5, 1.5)
 C(5.0, 5.0)
 D(3.0, 4.0)
 E(4.0, 4.0)
 F(3.0, 3.5)

Goal : 3 Clusters

Manhattan Distance Matrix

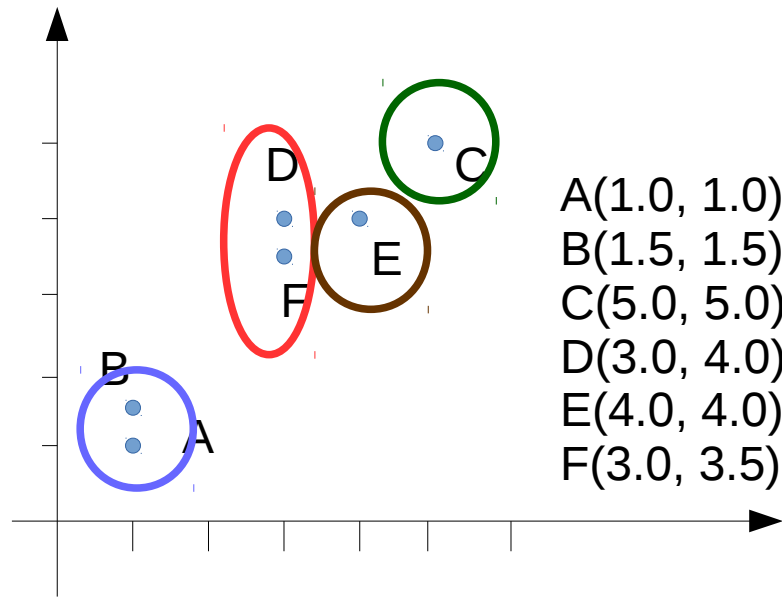
	A	B	C	D	E
A	0				
B	1	0			
C	8	7	0		
D	5	4	3	0	
E	6	5	2	1	0
F	4.5	3.5	3.5	0.5	1.5

	C	D	E	F
DistAvg	2.83	1.5	1.5	1.83
Dist(AB)	7	4	5	3.5

Max



Divisive Hierarchical Clustering (10)



Manhattan Distance Matrix

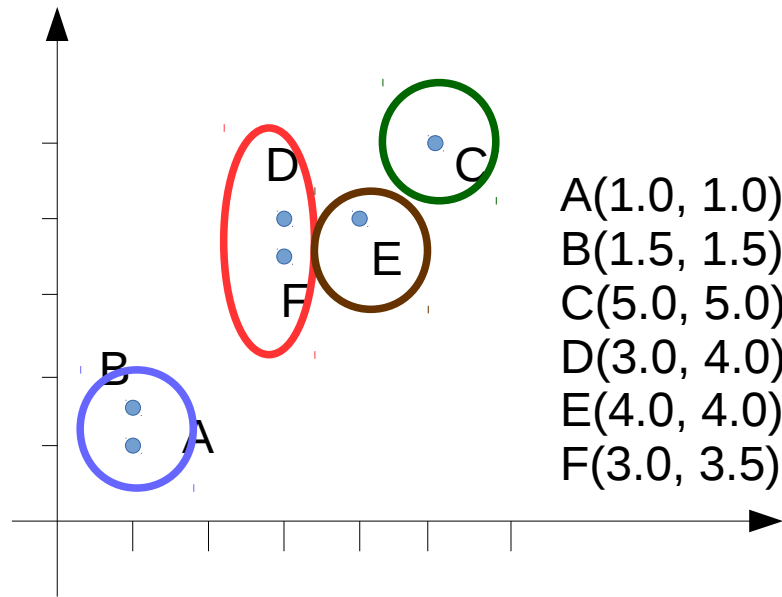
	A	B	C	D	E
A	0				
B	1	0			
C	8	7	0		
D	5	4	3		
E	6	5	2	1	
F	4.5	3.5	3.5	0.5	1.5

	D	E	F
DistAvg	0.75	1.25	1
Dist(AB)	4	5	3.5
Dist(C)	3	2	3.5

Continue the procedure ...



Divisive Hierarchical Clustering (11)



Keep the same clusters

Manhattan Distance Matrix

	A	B	C	D	E
A	0				
B	1	0			
C	8	7	0		
D	5	4	3		
E	6	5	2	1	
F	4.5	3.5	3.5	0.5	1.5

	D	F
DistAvg	0.5	0.5
Dist(AB)	4	3.5
Dist(C)	3	3.5
Dist(E)	1	1.5



Normalization

Min-max feature scaling :

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Standard score :

$$\frac{X - \mu}{\sigma}$$



Exercises

Use

- 1) Jaccard similarity coefficient
 - 2) Jaccard weighted coefficient
 - 3) Angular distance
- to analyze the following data

	Sugar (mg)	S1	S2	S3
Patient1	1	1	0	1
Patient2	2	0	1	1



Exercises

Use

- 1) Jaccard similarity coefficient
- 2) Jaccard weighted coefficient
- 3) Angular distance

to analyze the following data

	Salt (mg)	S1	S2	S3
Patient1	2	1	0	1
Patient2	3	0	1	1



Exercises

Let $Set2D = \{\mathbf{A}(2, 2), \mathbf{B}(3,3), \mathbf{C}(10, 10), \mathbf{D}(5, 8),$
 $\mathbf{E}(8,9), \mathbf{F}(5,7), \mathbf{G}(1,2)\}$ be a data set of 7 points.

Using Manhattan Distance to find 3 clusters with following methods:

- 1) K-means with the initial points : **A**, **B**, and C.
- 2) Agglomerative clustering with single linkage.
- 3) Agglomerative clustering with complete linkage.
- 4) Agglomerative clustering with average linkage.

